

National University of Modern Languages



Assignment # 1

Subject: IICT

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Section: ___A___

Assignment # 1

Q no 1: Explain the register and types of register.

REGISTER

Registers are a type of computer memory used to quickly accept, store, and transfer data and instructions that are being used immediately by the CPU. The registers used by the CPU are often termed Processor registers.

The computer needs processor registers for manipulating data and a register for holding a memory address. The register holding the memory location is used to calculate the address of the next instruction after the execution of the current instruction is completed.

TYPES OF REGISTER

- AC (accumulator)
- DR (Data registers)
- AR (Address registers)
- PC (Program counter)
- MDR (Memory data registers)
- IR (index registers)
- MBR (Memory buffer registers)

AC (accumulator)

An accumulator is a type of register included in a CPU. It acts as a temporary storage location that holds an intermediate value in mathematical and logical calculations. Intermediate results of an operation are progressively written to the accumulator, overwriting the previous value.

For example, in the operation " $3 + 4 + 5$," the accumulator would hold the value 3, then the value 7, then the value 12. The benefit of an accumulator is that it does not need to be explicitly referenced, which conserves data in the operation statement.

In modern CPUs, accumulators are replaced by general-purpose registers because they offer more flexibility. However, accumulators may still be in some special-purpose processors.

DR (Data registers)

A register used for holding information (either program words or data words) that is in the process of being transferred from the memory the central processor, or vice versa.

AR (Address registers)

Address location of memory is stored in this register to be accessed later. It is called by both MAR and MDR together.

PC (Program counter)

These registers are utilized in keeping the record of a program that is being executed or under execution. These registers consist of the memory address of the next instruction to be fetched.

MDR (Memory data registers)

All the information that is supposed to be written or the information that is supposed to be read from a certain memory address is stored here.

IR (index registers)

The instruction register (IR) is used to hold the instruction that is currently being executed. The contents of IR are available to the control unit, which generate the timing signals that control the various processing elements involved in executing the instruction.

MBR (Memory buffer registers)

Buffer registers are **a type of registers used to store a binary word**. These can be constructed using a series of flip-flops as each flip-flop can store a single bit. This means that in order to store an n-bit binary word one should design an array of n flip-flops